AI IN CASE STUDIES

AI-ENABLED INCIDENT DEMAND FORECASTING

Municipal Emergency Response — ERP-Integrated Decision Support

A 2024 911 Serviced Calls Case Study

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PRESENTATION OVERVIEW

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CASE STUDY TOPIC & BUSINESS PROBLEM

The operational challenge facing municipal emergency response

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2024 911 serviced calls data and analytical scope

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Key KPIs and the current performance picture

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BUSINESS VALUE, RISKS & NEXT STEPS

Expected impact, governance, and the path forward

SECTION 1

BUSINESS CONTEXT AND PROBLEM

Municipal emergency response operations are highly **time-sensitive** and **resource-constrained**.

1

REACTIVE DISPATCH

Dispatch decisions are often made in reaction to demand rather than in anticipation of it, limiting preparedness and response quality.

2

DEMAND VARIABILITY

Call volume varies significantly by hour, day, and location — yet resource positioning often does not reflect this variation.

3

CLUSTERED SLA BREACHES

Service level breaches tend to concentrate in specific precincts and shifts, suggesting structural rather than random failure patterns.

Core Problem: How can we move from reactive dispatch to AI-enabled, data-supported decision-making?

SECTION 2

CASE OBJECTIVE

This case study aims to design a realistic **AI-enabled decision-support framework** for 911 operations.

DATA-DRIVEN BASELINE

Establish a rigorous baseline using 2024 serviced 911 calls to ground all findings in real operational data.

PATTERN IDENTIFICATION

Identify demand patterns and performance bottlenecks, and detect where SLA breaches concentrate.

PREDICTIVE & PRESCRIPTIVE AI

Propose layered AI capabilities — from demand forecasting to resource allocation recommendations.

ERP & RPA INTEGRATION

Connect the AI solution to enterprise workflows and automation to drive actionable execution.

BUSINESS VALUE

Demonstrate operational and strategic value, supporting the case for digital transformation in emergency response.

SECTION 3

DATASET OVERVIEW

DATASET: POLICE_SERVICED_911_CALLS_2024.CSV

~324,871 RECORDS

Full-year 2024 coverage of serviced 911 calls

~26 COLUMNS

Rich multi-dimensional operational data fields

This gives a strong base for **descriptive**, **diagnostic**, and future **predictive** analysis.

KEY DATA FIELDS INCLUDED

- `called_at` — timestamp of call receipt
- `priority` — incident priority level
- `precinct` — geographic operational unit
- `category` — incident classification
- `call_description` — narrative description
- `total_response_time` — end-to-end response duration
- `intake_time` — call processing time
- `dispatch_time` — time to unit dispatch
- `travel_time` — unit travel to scene
- `on_scene_time` — time spent at incident
- Geographic fields for spatial analysis

SECTION 4

BASELINE PERFORMANCE SUMMARY

This baseline establishes the performance picture **before** proposing any AI intervention.

324,871

TOTAL CALLS ANALYZED

Full-year 2024 serviced call records

15.4M

MEDIAN RESPONSE TIME

Median total response time across all calls

95.3M

P90 RESPONSE TIME

The long delay tail affecting a significant share of cases

75.85%

SLA COMPLIANCE

Overall rate — meaning nearly 1 in 4 calls breach SLA

Key insight: The median looks manageable, but the P90 reveals a long delay tail. A significant share of cases take far longer than the central tendency suggests — on-scene time is the largest median component of the response chain.

SECTION 4 – BASELINE EVIDENCE

BASELINE PERFORMANCE: KPI SUMMARY

This gives the baseline performance picture before any AI intervention.

Performance by Priority

Priority	Calls	SLA Threshold	Median TRT	P90 TRT	SLA Compliance (%)
3	106300	75.0	28.4	199.3	75.27
2	104360	40.0	19.5	88.0	73.35
3	103030	15.0	9.7	20.0	79.02
5	3384	80.0	28.5	184.94	75.34
4	3442	120.0	39.95	289.99	75.49
Unknown	106	120.0	7.15	21.15	97.17

Priority 3 calls (the largest volume group) show the lowest SLA compliance at 75.27% — a key pressure point for the AI framework to address.

SECTION 5

DEMAND PATTERNS BY TIME

Call demand is **not flat**. It varies substantially over time — which directly justifies the forecasting logic at the core of this framework.

HOURLY DEMAND PATTERN

- Demand falls in the early morning hours
- Volume rises steadily through the day
- Strongest peak appears in the **late afternoon and early evening**

Chart reference: Demand_ByHour.png

MONTHLY VARIATION

- Meaningful variation exists across months of the year
- Weekly variation is present but less dramatic than hourly fluctuation

Chart reference: Demand_ByMonth.png

DAY-OF-WEEK DISTRIBUTION

- Weekends are slightly higher than mid-week
- **Saturday** appears to be the highest-volume day
- Wednesday and Thursday are relatively lower
- Variation is not extreme but large enough to matter for staffing

Chart reference: Demand_ByDayOfWeek.png

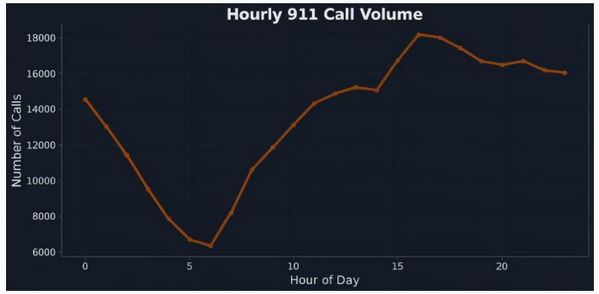
Key takeaway: This evidence supports demand forecasting rather than purely reactive dispatch. Even moderate day-of-week variation matters when combined with hourly surges and spatial concentration.

SECTION 5 – DEMAND EVIDENCE

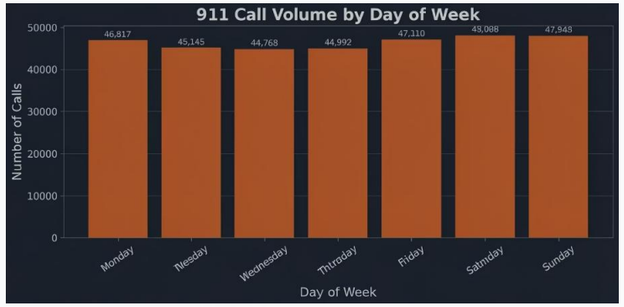
DEMAND PATTERNS: HOURLY, DAILY & MONTHLY

Demand is clearly not flat. It changes by hour and also across larger time patterns, which supports the need for forecasting rather than reactive dispatch.

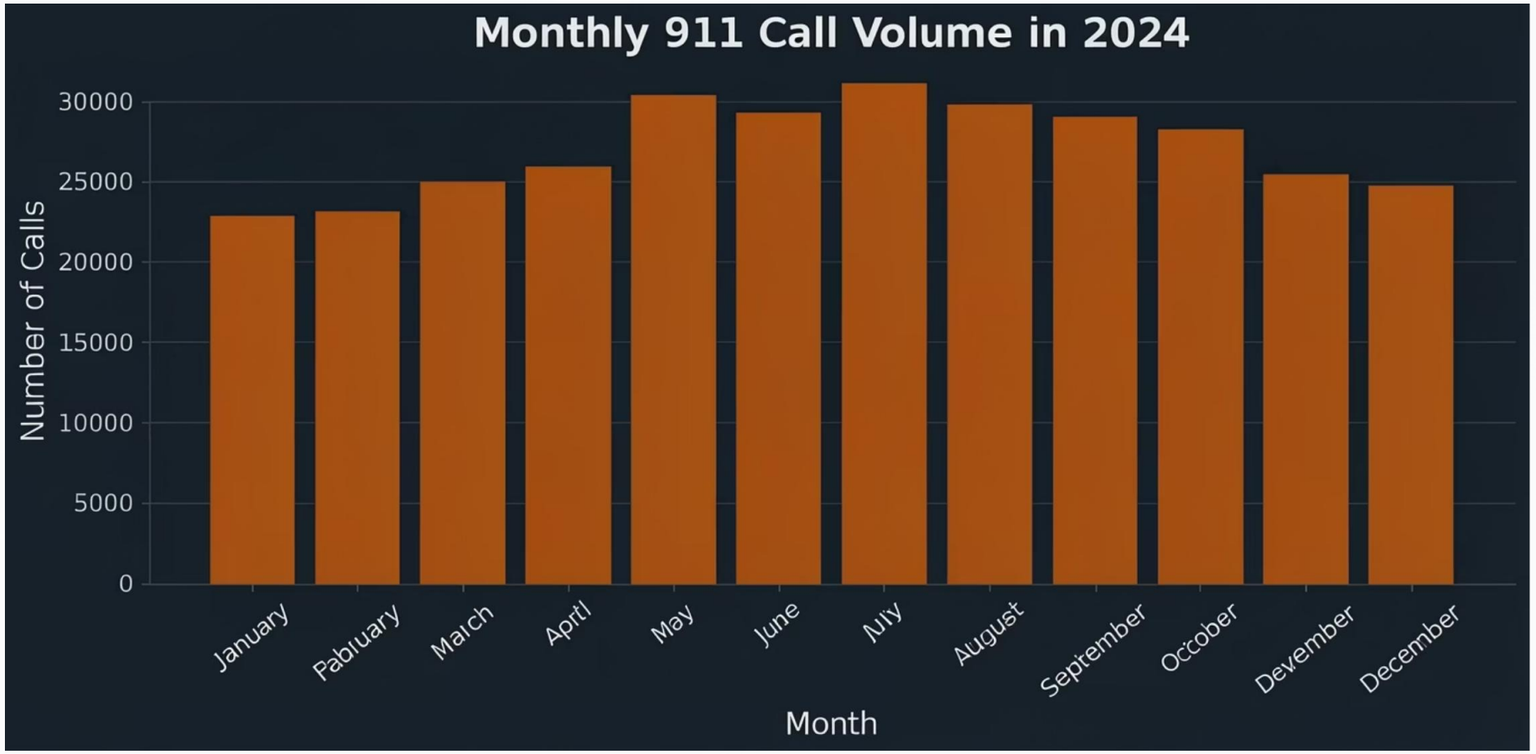
HOURLY CALL VOLUME



CALL VOLUME BY DAY OF WEEK



MONTHLY CALL VOLUME (2024)



SECTION 6 & 7

DEMAND CONCENTRATION BY PRECINCT AND SHIFT

Demand is not only temporal. It is also **spatially concentrated** — and that spatial pattern is operationally significant.

EVENING SHIFT DOMINANCE

The evening shift generally carries the **highest call volume** across precincts — creating predictable surge windows.

PRECINCT CONCENTRATION

Some precincts consistently handle much higher volume than others. Pressure is **not equally distributed** across the operating environment.

ZONE-BASED CASE FOR AI

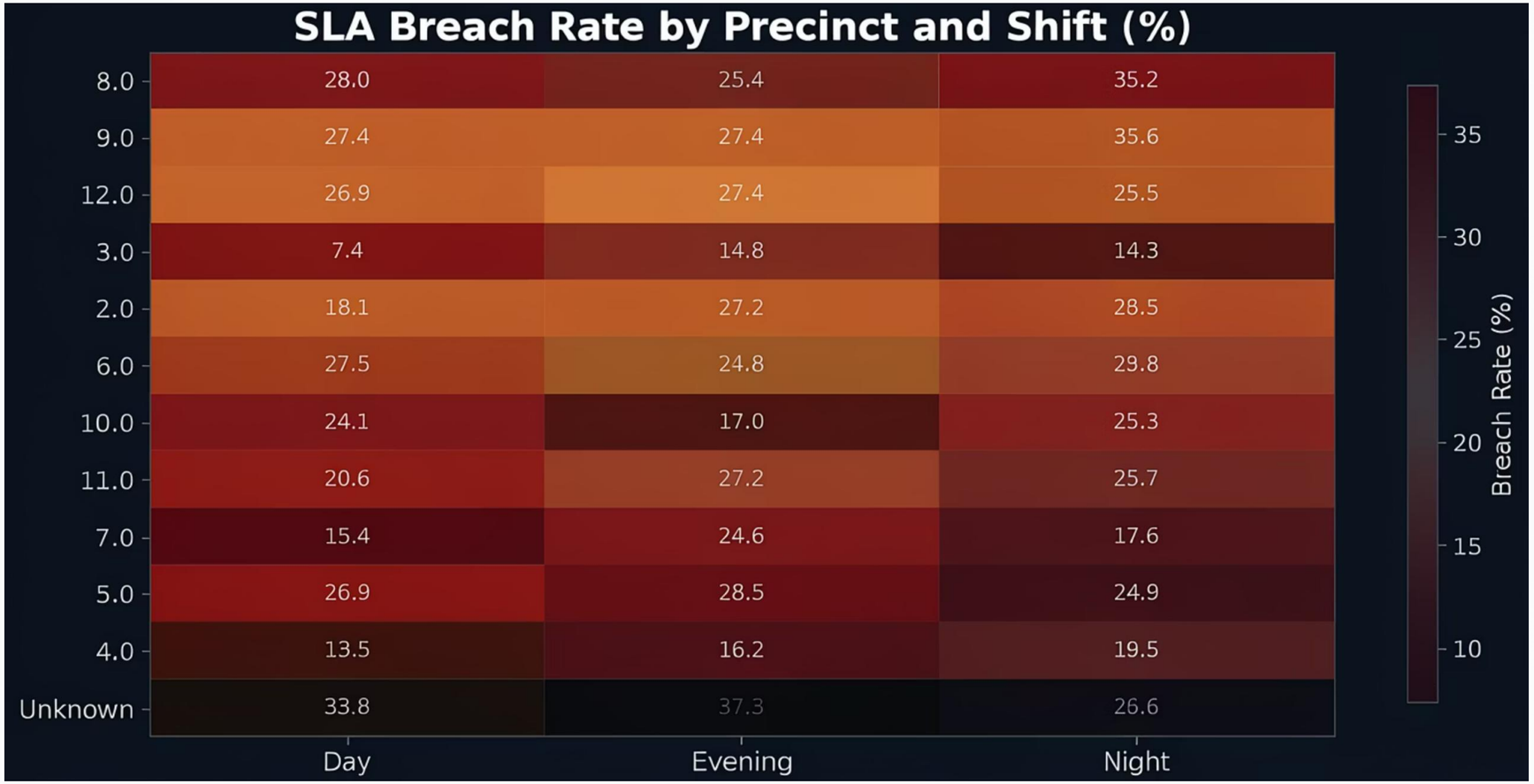
This concentration creates a strong case for **zone-based demand forecasting** and targeted allocation at the precinct-shift level.

Key takeaway: The operational challenge is not only "how many calls" — it is "where and when pressure accumulates."
Chart reference: Heatmap_Demand_Precinct_Shift.png

SECTION 6 – DIAGNOSTIC PRESSURE POINTS

SLA BREACH CONCENTRATION: PRECINCT × SHIFT HEATMAP

This helps locate where service pressure is concentrated — not just overall, but by operating area and shift.

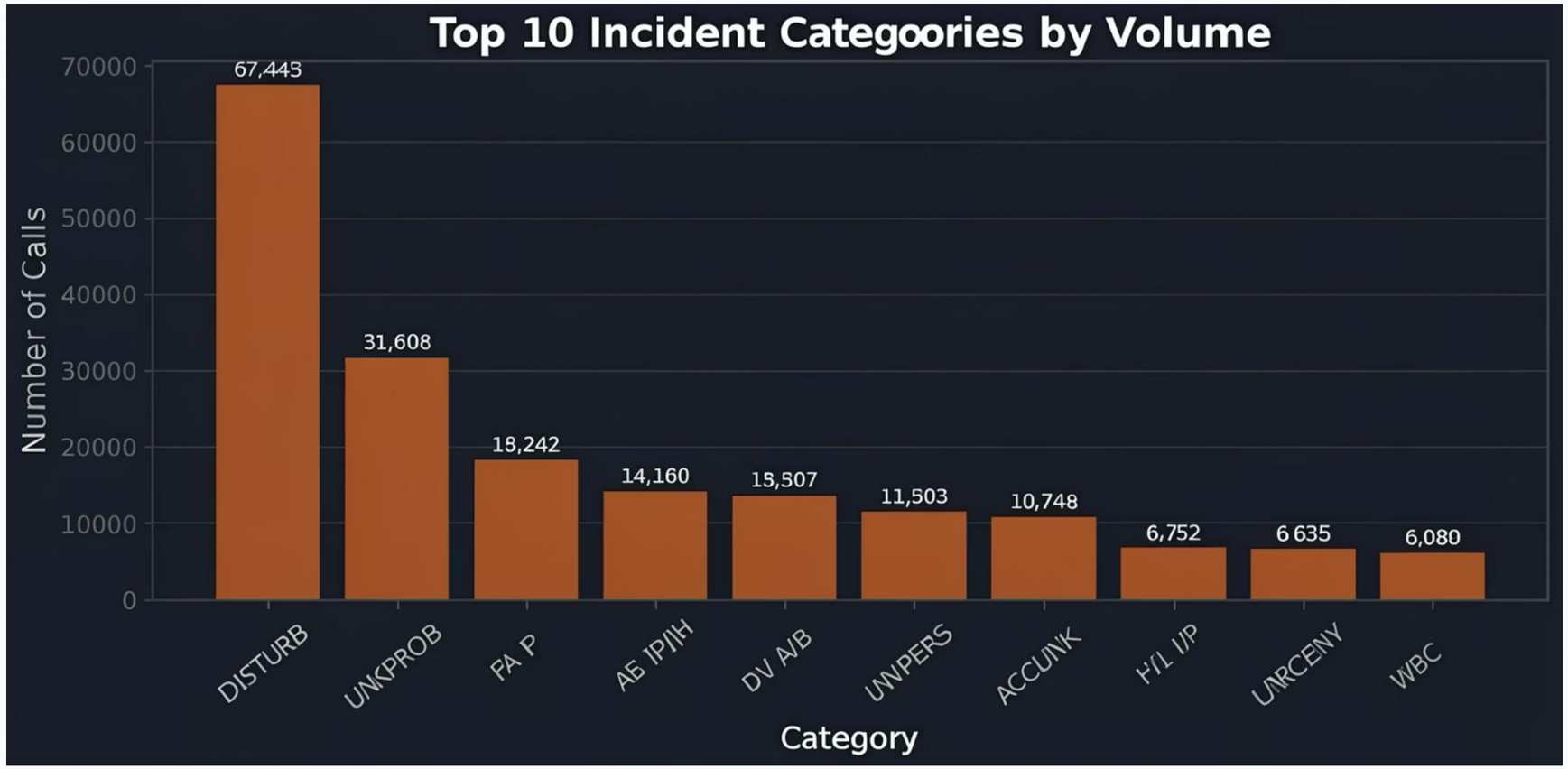


Precinct 8.0 Night shift shows the highest breach rate at 35.2%. Evening shifts consistently show elevated breach rates across multiple precincts — a structural pattern, not random noise.

SECTION 7 – OPERATIONAL COMPOSITION

TOP INCIDENT CATEGORIES BY CALL VOLUME

This shows which call types dominate workload and therefore shape resource pressure.



DISTURB (disturbance) calls alone account for 67,443 incidents — more than double the next category. Forecasting these high-volume types is essential for accurate resource planning.

SECTION 8

SLA BREACH CONCENTRATION BY PRECINCT AND SHIFT

This analysis is one of the **most important findings** in the case — breach rates are not uniformly distributed, they are structurally concentrated.

KEY OBSERVATIONS

- **Breach rates vary significantly** across precincts — some areas face structurally higher risk than others.
- **Night and evening shifts** show strong breach pressure across multiple precinct areas simultaneously.
- Some precincts face **consistently elevated breach rates** regardless of shift — indicating persistent resource gaps.
- This concentration indicates a **real opportunity for targeted AI intervention** in high-risk precinct–shift combinations.

STRATEGIC IMPLICATION

The problem is **not evenly spread**. A generalized solution will underperform.

AI support should be focused and prioritized on **high-risk precinct–shift combinations** where breach probability is highest and intervention value is greatest.

Chart reference: Heatmap_BreachRate_Precinct_Shift.png

SECTION 9

RESPONSE-TIME DECOMPOSITION

To improve response performance, we must first understand **where delay is actually occurring** in the response chain.

COMPONENT BREAKDOWN (MEDIAN)

ON-SCENE TIME

Largest median component. Field-side operational realities are the primary driver of total response duration.

TRAVEL TIME

Second largest component. Unit positioning and routing efficiency directly impact this segment.

DISPATCH TIME

Smaller but still **operationally relevant** — AI-supported prioritization can reduce decision latency here.

INTAKE TIME

Comparatively low. Call-taking is not the primary bottleneck in the current system.

Chart reference: Decomposition_ComponentMedians.png

Key takeaway: Delays are not driven primarily by intake or call-taking. Field-side timing and on-scene operational realities are the major contributors — which means improvement strategies must address resource positioning, not just call processing.

WORKLOAD COMPOSITION BY CATEGORY

A small number of dominant incident categories shape the majority of total workload. This means:

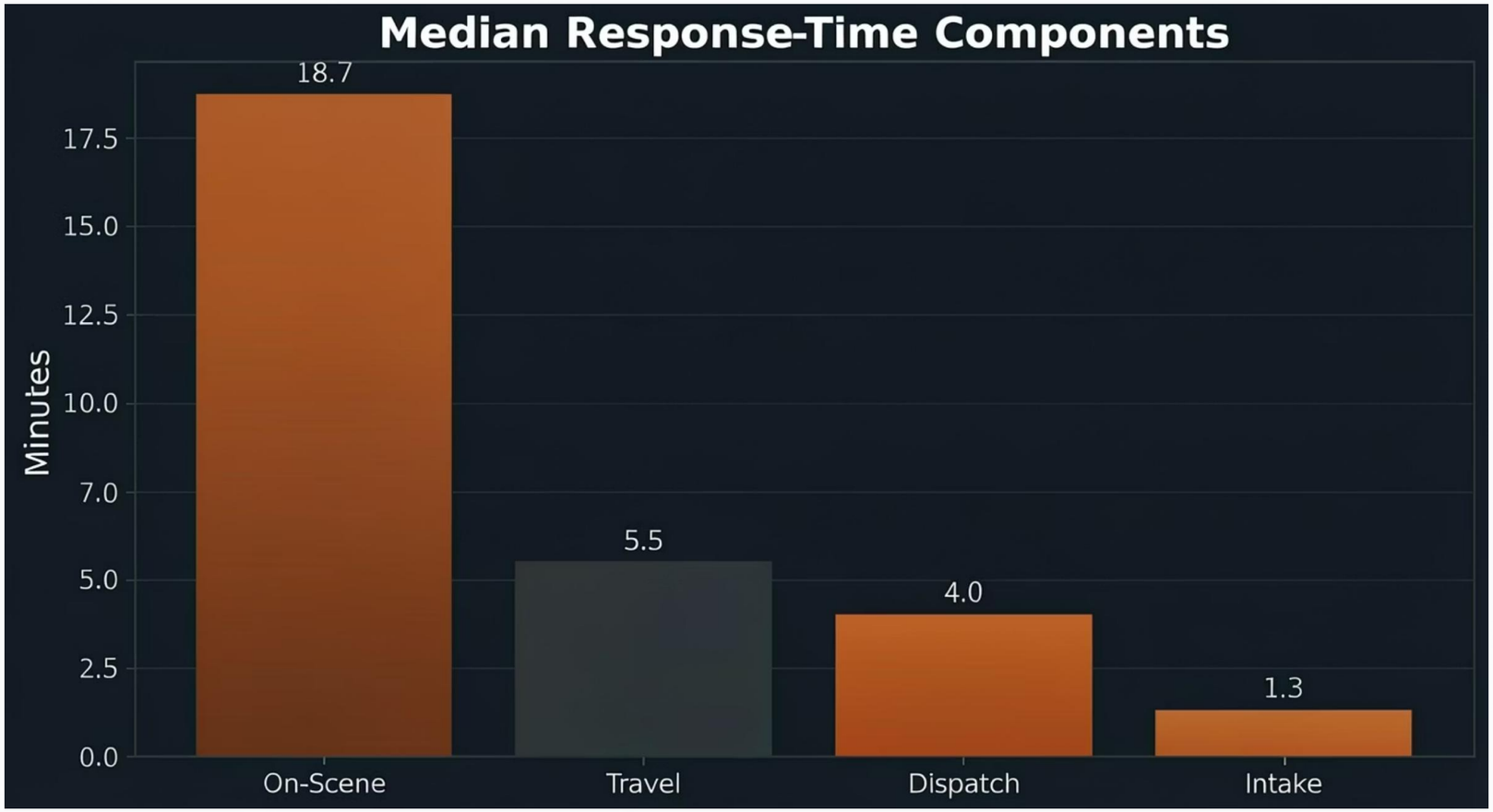
- Staffing and response planning cannot treat all calls as operationally equal
- Workload **composition** matters, not only total call count
- A second layer of analysis can examine which categories are most likely to breach SLA

Chart reference: TopCategories_ByVolume.png

SECTION 9 – BOTTLENECK ANALYSIS

RESPONSE-TIME DECOMPOSITION: WHERE IS THE DELAY?

This helps separate where the delay is occurring — inside intake/dispatch, or later in the field response chain.



On-Scene time (18.7 min median) dominates total response time — dwarfing Travel (5.5), Dispatch (4.0), and Intake (1.3). This shifts the optimization focus toward field operations and unit availability, not just dispatch speed.

SECTION 11

PROPOSED AI-ENABLED SOLUTION FRAMEWORK

The proposed solution is a **layered decision-support framework** — not a single model, but an integrated architecture of analytical and operational capabilities.

DESCRIPTIVE & DIAGNOSTIC

Understand current performance, demand patterns, and breach concentration

PREDICTIVE LAYER

Forecast demand by precinct and shift; predict SLA breach probability

PRESCRIPTIVE LAYER

Recommend resource allocation, dispatch prioritization, and unit positioning

ERP + RPA EXECUTION

Connect recommendations to workforce scheduling, fleet, and automated workflows

Operational logic: (1) Understand current performance → (2) Predict upcoming pressure → (3) Recommend feasible action → (4) Support execution through enterprise workflows.

SECTION 12

PREDICTIVE AND PRESCRIPTIVE AI LAYERS

PREDICTIVE LAYER

- Forecast demand by precinct and shift using historical call patterns
- Predict probability of SLA breach before pressure fully materializes

PRESCRIPTIVE LAYER

- Recommend resource allocation under operational constraints
- Support dispatch prioritization and unit pre-positioning
- Improve preparedness before surge conditions arrive

Business logic: The goal is not to replace dispatchers. The goal is to support them with better foresight and more structured, data-backed recommendations.

SECTION 13

ERP AND RPA INTEGRATION

The proposed framework is not only analytical. It is **enterprise-oriented** — designed to connect AI outputs to real operational execution.

ERP INTEGRATION

HR / WORKFORCE SCHEDULING

Demand forecasts feed directly into shift planning and staffing decisions across precincts.

FLEET & ASSET READINESS

Predicted surge windows trigger proactive vehicle and equipment readiness checks.

PROCUREMENT & INVENTORY

Resource consumption patterns inform supply chain and inventory replenishment triggers.

FINANCE & COST LOGIC

Allocation decisions are connected to cost tracking, overtime monitoring, and budget visibility.

RPA SUPPORT

AUTOMATED REPORTING

Routine performance and compliance reports generated and distributed without manual effort.

SLA EXCEPTION WORKFLOWS

Breach events automatically trigger escalation and exception-handling workflows.

WORKFLOW NOTIFICATIONS

Supervisors and dispatchers receive timely alerts when predicted risk thresholds are crossed.

ROUTINE OPERATIONAL TRIGGERS

Repetitive administrative tasks are automated, freeing staff for higher-value decision-making.

Key message: AI becomes more useful when it is connected to execution and enterprise processes — not siloed as a standalone analytical tool.

SECTION 14

BUSINESS VALUE AND EXPECTED IMPACT

The framework delivers value at both the **operational** and **strategic** levels of the organization.

OPERATIONAL VALUE

DEMAND VISIBILITY

Better visibility into upcoming demand surges before they fully materialize

BALANCED ALLOCATION

More balanced resource distribution across precincts and shifts

FEWER SLA BREACHES

Lower concentration of breaches in high-risk precinct–shift combinations

DECISION TRANSPARENCY

Improved auditability and justification for allocation decisions

STRATEGIC VALUE

SERVICE CONSISTENCY

More reliable and equitable emergency response across all precincts

MANAGERIAL CONTROL

Stronger data-backed oversight and accountability at the supervisory level

SCALABLE TRANSFORMATION

A replicable path toward digital transformation in emergency response operations

SECTION 15

RISKS, CONTROLS, AND GOVERNANCE

A responsible AI system must address accountability, fairness, and security — not just analytical performance.

PRIVACY & DATA MINIMIZATION

Only operationally necessary data fields are used. Personal identifiers are excluded or anonymized in line with data minimization principles.

ACCESS CONTROL & AUDIT LOGGING

All system interactions are logged. Role-based access controls ensure that only authorized personnel can view or act on AI recommendations.

FAIRNESS ACROSS PRECINCTS

Model outputs are monitored for systematic bias across precincts and shifts to ensure equitable resource distribution.

HUMAN-IN-THE-LOOP

Dispatchers and supervisors retain final decision authority. AI provides recommendations — it does not replace human judgment.

SECURITY & GOVERNANCE

Security controls are aligned with good governance practice. Empirical findings are clearly distinguished from simulated business assumptions.

Key principle: This is not automation without accountability. It is decision support with governance.

SECTION 16

WHAT HAS BEEN ACCOMPLISHED SO FAR

The case has moved from **concept stage** into **evidence-backed analytical groundwork**. The following work has been completed:

1 TOPIC, SCOPE & DATASET

Topic and scope definition, dataset selection and validation, and baseline KPI generation with SLA logic defined.

2 DEMAND ANALYSIS

Demand analyzed by hour, day, month, precinct, and shift — revealing clear temporal and spatial concentration patterns.

3 DIAGNOSTIC OUTPUTS

Breach heatmap generation, category-level workload analysis, and response-time decomposition across all components.

4 FRAMEWORK DESIGN

Full design of the AI + ERP + RPA framework, connecting predictive and prescriptive layers to enterprise execution workflows.

SECTION 17

NEXT STEPS

The next phase of the case will move from analytical groundwork into **model development, validation, and business integration**.

1 PREDICTIVE MODEL DEVELOPMENT

Train and validate demand forecasting models and breach-risk prediction logic at the precinct–shift level.

2 PRESCRIPTIVE ALLOCATION SCENARIOS

Build and test prescriptive resource allocation scenarios under realistic operational constraints.

3 ERP ASSUMPTIONS & COST LOGIC

Refine ERP integration assumptions and strengthen business impact estimation with cost and efficiency logic.

4 FINAL INTEGRATION & PRESENTATION

Integrate final figures, validated findings, and prescriptive outputs into the article and final presentation deliverable.

SECTION 18

CONCLUSION

This case study shows that **municipal 911 operations** provide a strong real-world setting for applying AI in business decision-making.

DEMAND IS VARIABLE

Call volume fluctuates meaningfully by hour, day, month, and location — making static resource planning insufficient.

PRESSURE IS CONCENTRATED

Specific precinct–shift combinations consistently carry disproportionate operational load.

BREACHES ARE STRUCTURAL

SLA failures are not randomly distributed — they cluster in predictable, addressable patterns.

DATA SUPPORTS ACTION

Real operational data provides a credible foundation for predictive, prescriptive, and enterprise-integrated decision support.

Final message: The goal is not just smarter analytics. The goal is better operational decisions, better allocation, and better service performance — for every precinct, every shift, every call.

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THANK YOU

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